



PORTOFOLIO | DATA MINING & MACHINE LEARNING

Predictive Modeling for Chronic Kidney Disease (CKD)

Performance and Interpretability Comparison
of LASSO, Decision Tree, and Neural Network

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Paradox Modelling Clinics: Accuracy vs. Transparency



Latar Belakang

Diagnosis CKD mengandalkan kombinasi indikator fungsi ginjal, blood parameters, and komorbidities.



Thallenge

Highly accurate algorithms are activated (such as Neural Networks) are often Black-Box, while clinical clinical decisions require transparency (White-Box).



Goal Analysis

Evaluate 3 algorithms (LASSO Logistic Regression, Decision Tree, Neural Network) to find the optimal balance between discrimination ability, calibration, and ease of clinical interpretation klinik.

Data Architecture: From Raw Data to Validation Set

1. Raw Data

CKD Dataset

(400 Observations)

- 250 CKD, 150 Not CKD
- High missing values in rbc (38%) and rc (32.8%)

2. Preprocessing

- **Imputation:** Median (Numeric), Mode (Categorical)
- **Transformation:** One-Hot Encoding
- **Standardization** (Especially for LASSO & NN)

Development Set

300 Observations

CKD Proportion: 62.7%

Stratified Split (75:25)

Internal Validation Set

100 Observations

CKD Proportion: 62.0%

Model Development Strategy

1. Cross-Validation

Stratified 3-fold Cross-Validation on the Development Set for stability of estimation.

2. Tuning Metric

Hyperparameter optimization based on **AUC (Area Under the ROC Curve)** to maximize discrimination capability at various thresholds.

3. Final Evaluation

Testing on the **Internal Validation Set** (Threshold 0.5) to measure discrimination (**AUC**) and probability calibration (**Brier Score**).

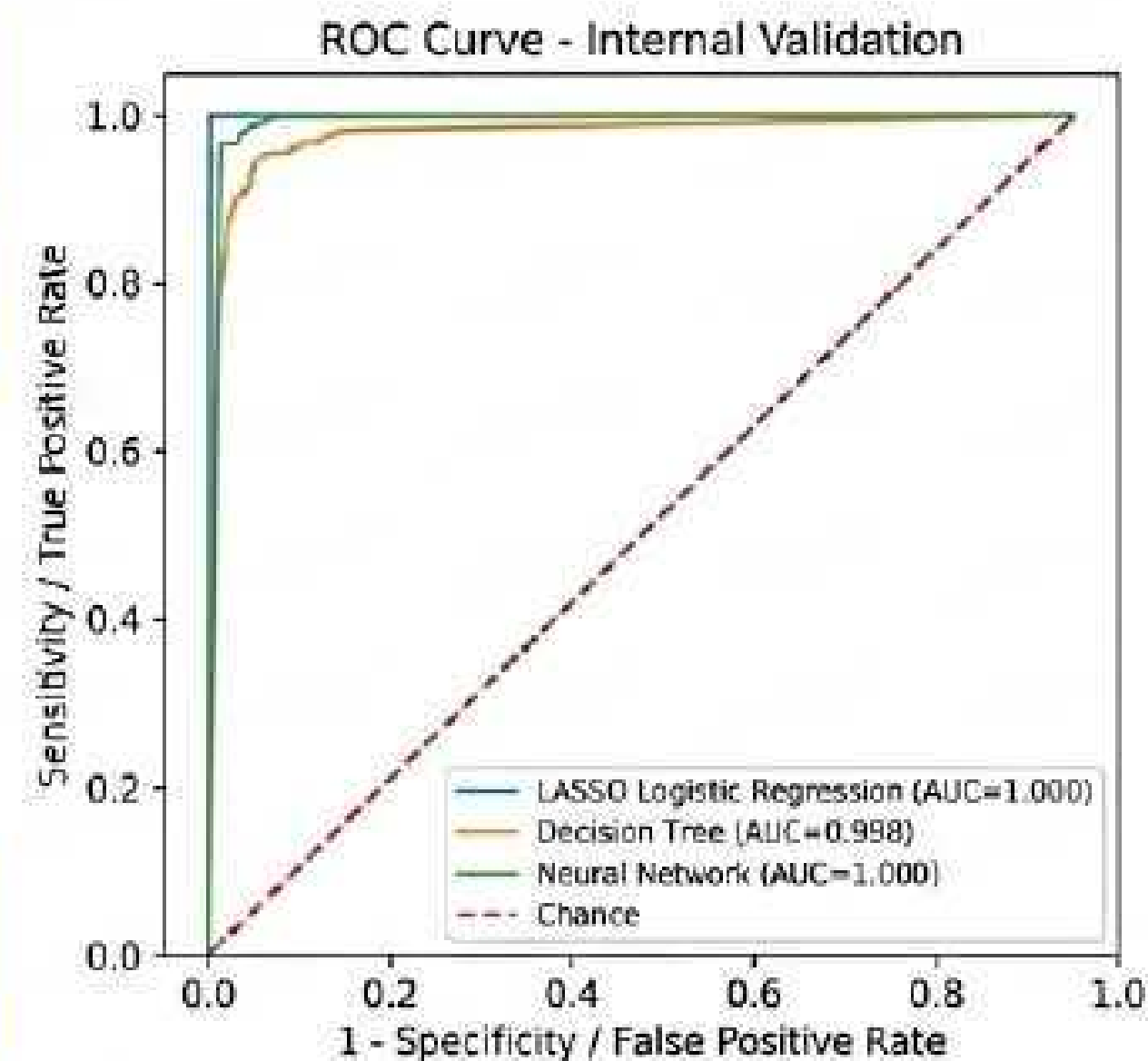
Initial Potential (Mean CV AUC):

LASSO (0.999) | Neural Network (0.991) | Decision Tree (0.982)

Moment of Truth: Internal Validation Results

Model	Accuracy	Sensitivity	Specificity	AUC	Brier Score
LASSO	1.000	1.000	1.000	1.000	0.0016
Decision Tree	0.990	1.000	0.974 (1 False Positive)	0.998	0.0124
Neural Network	0.980	0.968 (2 False Negatives)	1.000	1.000	0.0665

Conclusion: LASSO not only achieves absolute accuracy (100%), but also produces the best probabilistic calibration (lowest Brier Score).



Dissecting the LASSO Brain: Key CKD Predictors

L1 penalty extracts the most clinically relevant variables.

Pulling towards Not CKD / Protective

Specific Gravity (sg): -2.502 log-odds

Hemoglobin (hemo): -2.222 log-odds

Pushing towards CKD / Risk Factors

Albumin (al): +1.685 log-odds

Diabetes Mellitus (dm_yes): +1.553 log-odds

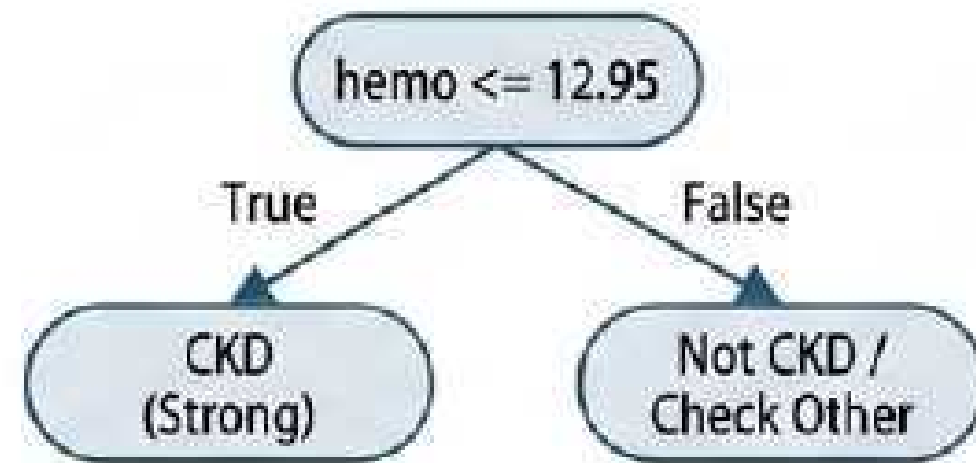
Appetite Poor (appet_poor): +1.531 log-odds

0

Clinical Insight: This mathematical pattern aligns with medical literature: changes in indicators of renal function (sg, al) and blood profile (hemo) are absolute predictors of CKD.

Alternative Models: Simplicity vs. Complexity

Decision Tree (Simplicity)



- Very concise rule, highly dependent on Hemoglobin (Importance 0.817).
- Main Rule: If hemo ≤ 12.95 , strong tendency towards CKD.
- Weakness: Too rigid, resulted in 1 False Positive.

Neural Network (Complexity)



- Able to capture non-linear patterns. Top predictors: su, hemo, sg.
- Weakness: Black-Box nature complicates extraction of individual patient risk scores.
- Resulted in 2 False Negatives and wide probability variation.

Synthesis: Why is LASSO the Winner?



Discrimination

LASSO and NN lead (AUC 1.000). Able to perfectly separate CKD and Not CKD classes on validation data.



Calibration

LASSO excels absolutely (Brier Score 0.0016). Prediction probabilities are very convincing and well-calibrated.



Clinical Transparency

LASSO reduces dimensions and produces log-odds coefficients that can be converted into a transparent score similar to a nomogram.

LASSO Logistic Regression is the ideal Blueprint: Mathematically robust, clinically transparent.

Conclusion & Clinical Implementation Roadmap

Main Conclusion

The clinical and laboratory characteristics of the CKD dataset have very strong predictive signals. LASSO Logistic Regression has proven to be the most balanced algorithm for mapping these signals.

Current Limitations

Perfect performance (100% Accuracy) is influenced by the limited sample size in the internal validation set and clear class separation.

Critical Recommendation: External Validation

Before implementation in a hospital Clinical Decision Support System (CDSS), the model **MUST** be tested on datasets from different populations, times, or demographics to ensure real-world parameter stability.